

Gour Gupal Talukder Shawon

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Research Interests

Multi-agent LLM systems, AI evaluation and benchmark design, federated learning and privacy-preserving machine learning, large language model alignment, natural language processing, software engineering automation, and distributed systems.

Education

Shahjalal University of Science and Technology

B.Sc. in Software Engineering CGPA: 3.45/4.00 (through 7th semester)

Sylhet, Bangladesh

Mar 2022 - May 2026

Research Experience

HiveMind Research Group

Research Contributor

Remote

Jan 2026 - Present

Paper: *AgentCollabBench: Diagnosing When Good Agents Make Bad Collaborators* Under Review (NeurIPS 2026)

Resources: [arXiv Preprint](#) | [GitHub](#) | [HuggingFace Dataset](#)

- Contributed to *AgentCollabBench*, a **benchmark for evaluating reliability and communication failures in multi-agent LLM systems** across Software Engineering, DevOps, and Data Engineering tasks.
- Assisted in early-stage **metric and dataset planning** for controlled multi-agent evaluation scenarios, including experiments related to **inter-agent influence** and **echo-chamber behavior**; created structured JSON benchmark samples and task configurations.
- Reviewed benchmark task structures, **communication topologies**, and annotation workflows to help improve **dataset consistency** and evaluation quality across collaborative agent pipelines.
- Participated in collaborative benchmark development and research review processes, contributing to **dataset validation**, experimental refinement, and technical coordination within a distributed AI research team.

Undergraduate Thesis

Commit Message Generation using LLMs with Issue-Based Context

May 2025 - Nov 2025

Supervisor: Mahfuzur Rahman Emon, Lecturer, IICT, SUST

- Conducted a **comparative study of GPT-4 Turbo, Claude 3.5, and Gemini 1.5 Pro** on a curated dataset of **309 real-world issue-linked commits** from high-authority repositories (freeCodeCamp, OWASP).
- Built a **dual evaluation framework** combining NLP metrics (**BLEU, BERTScore**) with double-blind assessment by **30 expert developers**, supported by a full-stack Human Evaluation Platform (Next.js, Node.js, MongoDB).
- Achieved **42–53% performance gain** in commit message quality over human-written baselines through **optimized prompt engineering**.

Advanced Machine Intelligence Research Lab, AIUB

Intern Research Assistant

Remote

Feb 2026 - Present

- Contributing to early-stage research on **federated unlearning for LoRA-tuned LLMs**, focusing on **privacy-aware machine learning** and **GDPR "Right to be Forgotten"** compliance in decentralized environments.
- Assisting with literature review, dataset preparation, and preliminary experimental analysis for research involving **non-IID federated learning**, statistical heterogeneity metrics, and **parameter-efficient fine-tuning** workflows.
- Participating in an ongoing **CVPR-oriented** research initiative on **spectral restoration of snow-degraded images** using **diffusion-based models**, including proposal development, dataset exploration, and baseline analysis.
- Collaborating within a small research team on experiment planning, technical discussions, and iterative research documentation under faculty supervision.

Industry Experience

ShellBeeHaken Ltd.

Intern Software Engineer (Onsite)

Dhaka, Bangladesh

Dec 2025 - May 2026

- Contributed to **production UI components** and navigation flows using **React Native (Expo)** and **Next.js**, integrating **REST APIs** and managing application state across screens.
- Assisted in building **event-driven backend services** with **AWS Lambda**; gained hands-on exposure to distributed systems concepts including **data replication** and **load balancing** during architecture discussions.
- Participated in the full **API development cycle** from writing data contracts to **integration testing**, while working in an **Agile** team with sprint planning and code reviews.

Technical Skills

Languages	JavaScript (ES6+), Java, C++, Python, TypeScript
Frontend	Next.js, React, React Native, Flutter
Backend	Node.js, Express.js, RESTful APIs, GraphQL, Socket.IO
Databases	MongoDB, Firebase, MySQL
Tools	AWS, Docker, Git, Linux, Serverless (AWS Lambda), Postman, Maven

Projects

Personality Detection Using Machine Learning

[GitHub](#) | [Live](#)

- Improved model **F1 score by 20%** via advanced **feature engineering** and **hyperparameter tuning**, achieving **91% accuracy** in real-time personality prediction.
- Tools: Python, scikit-learn, Flask, Next.js, Pandas, NumPy, joblib.

Local Household Service Provider Platform

[GitHub](#) | [Live](#)

- Designed an **optimized data-fetching architecture** that reduced retrieval time by **60%**, resulting in **20% higher** overall application responsiveness and user satisfaction.
- Integrated **Google Gemini-powered AI chatbot**, improving service query response rate by **60%**; authored critical system documents including SRS and project proposal.
- Tools: Next.js, Express.js, MongoDB, Nodemailer, Cloudinary.

Real-Time Multi-Module Chat Application

[GitHub](#)

- Built a **real-time desktop chat system** using **Java, Socket.IO (Netty), PostgreSQL, and Swing**, structured with a **multi-module Maven architecture** and shared DTOs for consistent client-server communication.
- Implemented **WebSocket-based messaging** with a custom **reconnection strategy (exponential backoff)** and heartbeat monitoring to ensure resilient real-time connectivity.
- Designed an **event-driven backend architecture** supporting authentication, messaging, and file transfer workflows, with secure and efficient persistence using **HikariCP, PostgreSQL indexing, and BCrypt authentication**.

Algorithmic Problem Solving

- Solved **670+** [LeetCode](#) problems; maximum streak of **433** days; peak contest rating **1622**; earned 23 badges.
- Completed 250–300 algorithmic problems across Codeforces, UVA, and VJudge platforms.

Academic Honors & Certifications

- **2nd Runner-up** - “Bridging Engineering & Computer Science” International Hackathon, organized by ILM-AI, co-hosted by **St. George’s, University of London**. Built a browser-based projectile motion simulator with optional air resistance (two-member team). [GitHub](#) | [Live](#)
- **Certifications:** AWS Cloud Practitioner · AWS Lambda · AWS DynamoDB · AWS API Gateway · Claude Code in Action (Anthropic) · Generative AI (Udacity & AWS) · ML Crash Course (Google).